

MATH4041/MCSC6120 Assignment 1

Glucose Control, Part 1

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1. Derivation of the Two-Degree-of-Freedom Model

The glucose control model introduced in lecture is

$$u = \alpha v + \beta \int_{-\infty}^t \lambda e^{-\lambda(t-t')} v(t') dt', \quad (1)$$

together with

$$\dot{v} = -\gamma v - \delta u(v + \bar{v}) + F(t). \quad (2)$$

To eliminate the integral, differentiate the first equation with respect to t :

$$\dot{u} = \alpha \dot{v} + \beta \lambda v - \beta \lambda \int_{-\infty}^t e^{-\lambda(t-t')} v(t') dt'. \quad (3)$$

From the original equation,

$$\beta \int_{-\infty}^t \lambda e^{-\lambda(t-t')} v(t') dt' = u - \alpha v. \quad (4)$$

Therefore,

$$\beta \lambda \int_{-\infty}^t e^{-\lambda(t-t')} v(t') dt' = \lambda(u - \alpha v). \quad (5)$$

Substituting this into the differentiated equation gives

$$\dot{u} = \alpha \dot{v} + \beta \lambda v - \lambda(u - \alpha v) \quad (6)$$

$$= -\lambda u + \lambda(\alpha + \beta)v + \alpha \dot{v}. \quad (7)$$

Define

$$f(u, v) = -\gamma v - \delta u(v + \bar{v}) + F. \quad (8)$$

Since

$$\dot{v} = f(u, v), \quad (9)$$

the model can be written as the two-degree-of-freedom ODE system

$$\begin{aligned} \dot{u} &= -\lambda u + \lambda(\alpha + \beta)v + \alpha f(u, v), \\ \dot{v} &= -\gamma v - \delta u(v + \bar{v}) + F. \end{aligned}$$

 (10)

2. Scaling the Variables

Following the lecture, introduce the dimensionless variables

$$v = \bar{v} v', \quad u = \tilde{u} u', \quad t = \tilde{t} t'. \quad (11)$$

For the v -equation,

$$\frac{\bar{v}}{\bar{t}} \frac{dv'}{dt'} = -\gamma \bar{v} v' - \delta \tilde{u} \bar{v} u'(v' + 1) + F. \quad (12)$$

Dividing by $\bar{v}/\tilde{t}$ gives

$$\frac{dv'}{dt'} = -(\gamma \tilde{t}) v' - (\delta \tilde{t} \tilde{u}) u'(v' + 1) + \frac{\tilde{t}}{\bar{v}} F. \quad (13)$$

Choose the scales so that the coefficients of the first two terms are equal to one:

$$\gamma \tilde{t} = 1, \quad \delta \tilde{t} \tilde{u} = 1. \quad (14)$$

Hence,

$$\boxed{\tilde{t} = \frac{1}{\gamma}, \quad \tilde{u} = \frac{\gamma}{\delta}.} \quad (15)$$

Define the dimensionless input parameter

$$\boxed{G = \frac{\tilde{t}}{\bar{v}} F = \frac{F}{\gamma \bar{v}}.} \quad (16)$$

The v -equation therefore becomes

$$\frac{dv'}{dt'} = -v' - u'(v' + 1) + G. \quad (17)$$

Define

$$f'(u', v') = -v' - u'(v' + 1) + G. \quad (18)$$

For the u -equation,

$$\frac{\tilde{u}}{\bar{t}} \frac{du'}{dt'} = -\lambda \tilde{u} u' + \lambda(\alpha + \beta) \bar{v} v' + \alpha f(u, v). \quad (19)$$

Using

$$f(u, v) = \frac{\bar{v}}{\tilde{t}} f'(u', v'), \quad (20)$$

gives

$$\frac{du'}{dt'} = -(\lambda \tilde{t}) u' + \lambda \bar{v}(\alpha + \beta) \frac{\tilde{t}}{\tilde{u}} v' + \alpha \frac{\bar{v}}{\tilde{u}} f'(u', v'). \quad (21)$$

Using the scales above, define the dimensionless parameters

$$\boxed{a = \frac{\lambda}{\gamma}, \quad b = \frac{\lambda \bar{v} \delta}{\gamma^2}(\alpha + \beta), \quad c = \frac{\alpha \bar{v} \delta}{\gamma}.} \quad (22)$$

Dropping the primes gives the dimensionless system

$$\boxed{\begin{aligned} \dot{u} &= -au + bv + cf(u, v), \\ \dot{v} &= -v - u(v + 1) + G, \end{aligned}} \quad (23)$$

where

$$f(u, v) = -v - u(v + 1) + G. \quad (24)$$

3. Equilibrium Solutions

At equilibrium,

$$\dot{u} = \dot{v} = 0. \quad (25)$$

Since $f(u, v) = 0$ at equilibrium, the first equation gives

$$-au + bv = 0. \quad (26)$$

Therefore,

$$\boxed{u^* = \frac{b}{a}v^*}. \quad (27)$$

The second equilibrium condition is

$$-v - u(v + 1) + G = 0. \quad (28)$$

Substituting

$$u = \frac{b}{a}v \quad (29)$$

gives

$$-v - \frac{b}{a}v(v + 1) + G = 0. \quad (30)$$

Expanding,

$$-\frac{b}{a}v^2 - \frac{b}{a}v - v + G = 0. \quad (31)$$

Multiplying by $-a/b$ gives

$$v^2 + v + \frac{a}{b}v - \frac{a}{b}G = 0, \quad (32)$$

or

$$v^2 + \left(1 + \frac{a}{b}\right)v - \frac{a}{b}G = 0. \quad (33)$$

Thus, exactly as in the lecture,

$$\boxed{v^* = -\frac{1}{2}\left(1 + \frac{a}{b}\right) \pm \frac{1}{2}\sqrt{\left(1 + \frac{a}{b}\right)^2 + \frac{4aG}{b}}}. \quad (34)$$

The corresponding equilibrium values of u are

$$\boxed{u^* = \frac{b}{a}v^*}. \quad (35)$$

The two equilibrium solutions coincide when

$$\left(1 + \frac{a}{b}\right)^2 + \frac{4aG}{b} = 0. \quad (36)$$

Solving for G gives the saddle-node value

$$G_{\text{SN}} = -\frac{b}{4a} \left(1 + \frac{a}{b}\right)^2. \quad (37)$$

Therefore, for $a > 0$ and $b > 0$, there are two real equilibrium solutions when

$$G > G_{\text{SN}}, \quad (38)$$

one equilibrium at

$$G = G_{\text{SN}}, \quad (39)$$

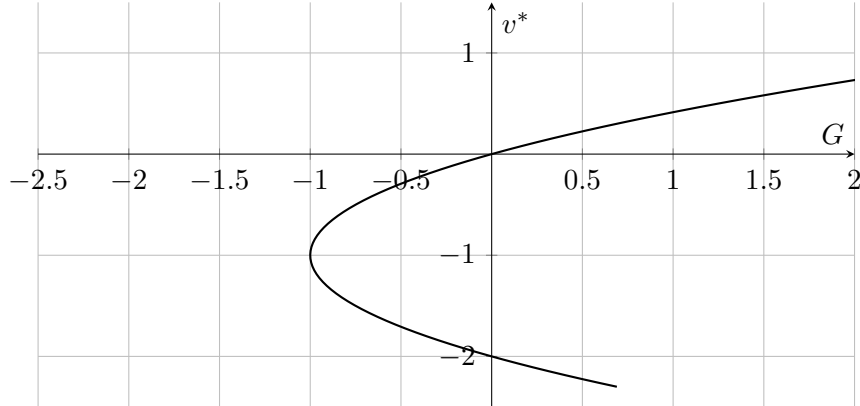
and no real equilibrium solutions when

$$G < G_{\text{SN}}. \quad (40)$$

Solving the equilibrium equation for G gives

$$G = v^* + \frac{b}{a} v^* (v^* + 1). \quad (41)$$

Thus, the equilibrium branch has the saddle-node shape shown in lecture. A representative sketch is shown below.



The figure illustrates the characteristic saddle-node shape of the equilibrium branch versus the input parameter G .